

Los Angeles Crime Analysis

ENGR 315

The Royal Dukes

Ryan Beach, Bodhi Theron, Arslan Malik

Task 1 - Dataset Selection

Our selected dataset is the [Crime Data from 2020 to 2024 in Los Angeles](#). This data was collected by utilizing the Los Angeles Police Department's crime and arrest data from their records management system.

The purpose of this collecting crime data is primarily to enhance law enforcement strategies to reduce crime, but also provide the public with current and accurate comprehensive data.

We particularly wanted to analyze how COVID-19 may have affected the amount of crime. This dataset is quite large with roughly 1 million data entries with lots of information including the type of crime, date, time of day, area, and victim age.

Task 2 - Question 1

- Question
 - How has the total and frequency of different types of crimes changed over time from 2020 to 2024?
- Why is this interesting to us?
 - This question is interesting because it will allow us view how the global pandemic may have affected crime rate and if particular types of crimes have increased or decreased since then.
- Why is it an inferential question?
 - This is an inferential question because we will have to use our knowledge of the time period and apply groupings and data analysis to conclude things such as the cause of increase/decrease of a crime since 2020.
- Potential Variables
 - Crime Code Description
 - Year reported
 - Crime Code

Task 2 - Question 2

- Question 2
 - How does the type of crime vary across different areas or neighborhoods in Los Angeles?
- Why is this interesting to us?
 - It can show whether certain areas experience more violent crime vs property crime, which has real-world implications for safety and policy.
- Why is it an inferential question?
 - This is an inferential question because we will analyze relationships between location and crime patterns, not just listing crimes in each area.
- Potential Variables
 - Area name / reporting district
 - Crime type
 - Frequency of incidents

Task 2 - Question 3

- Question 3
 - How does the time of day influence the frequency and type of crimes committed?
- Why is this interesting to us?
 - This is interesting as we have the suspicion that crime rate will increase at night and will be able to prove this by analyzing the dataset. In addition, analysis could reveal peak crime hours such as at night or daytime for certain types of crime.
- Why is it an inferential question?
 - This is an inferential question as it requires identifying patterns and correlations between time and behavior, as opposed to a raw count of time or types of crime.
- Potential Variables
 - Time of incident
 - Crime type
 - Number of crimes

Task 2 - Question 4

- Question 4
 - What is the relationship between weapon use and the severity or category of crimes in Los Angeles?
- Why is this interesting to us?
 - This is interesting as it will help us understand how weapons influence crime severity. This data could then be used to highlight risks and inform prevention strategies.
- Why is it an inferential question?
 - This is an inferential question as it examines a relationship between weapon presence and crime type/severity, rather than a simple count or yes/no question.
- Potential Variables
 - Weapon used (yes/no)
 - Crime category
 - Incident count

Task 3 - Data & Questions - Table

Question	Feature	Answerable
How has the frequency of crimes changed from 2020 to 2024?	Date OCC, Crime Code	Yes
How does crime vary by area?	Crime Code, Area	Yes
How does the time of day influence crime?	Time OCC, Crime Code	Yes
What is the relationship between weapon use and crime severity?	Weapon Used Code, Crime Code	Maybe

Task 3 - Question 1 Analysis

Frequency of different types of crimes over time

The image on the right demonstrates how the dataset will provide us with sufficient data to determine how the amount of different types of crimes have increased or decreased over time, as it provides data for categories such as the date of the incident, and the type of incident.

From the analysis, we expect to find the relationship between a crime and how the frequency of the event has changed since the pandemic.

DATE OCC	TIME OCC	AREA	AREA NAM	Rpt Dist	Nc Part 1-2	Crms Cd	Crms Cd Desc
11/7/2020 0:00	845	15	N Hollywo	1502	2	354	THEFT OF IDE
10/18/2020 0:00	1845	15	N Hollywo	1521	1	230	ASSAULT WIT
10/30/2020 0:00	1240	9	Van Nuys	933	2	354	THEFT OF IDE
12/24/2020 0:00	1310	7	Wilshire	782	1	331	THEFT FROM
9/29/2020 0:00	1830	14	Pacific	1454	1	420	THEFT FROM
11/11/2020 0:00	1210	4	Hollenbec	429	2	354	THEFT OF IDE
4/16/2020 0:00	1350	3	Southwest	396	2	354	THEFT OF IDE
7/7/2020 0:00	1400	11	Northeast	1133	2	812	CRMAGNST C
3/2/2020 0:00	1200	17	Devonshire	1729	2	354	THEFT OF IDE
9/1/2020 0:00	900	21	Topanga	2196	2	354	THEFT OF IDE
6/20/2020 0:00	1	6	Hollywood	657	2	812	CRMAGNST C
11/17/2020 0:00	1320	20	Olympic	2023	2	354	THEFT OF IDE
9/5/2020 0:00	1500	21	Topanga	2149	1	510	VEHICLE - STO
7/2/2020 0:00	500	17	Devonshire	1762	1	310	BURGLARY
2/13/2020 0:00	2300	14	Pacific	1406	1	330	BURGLARY FF
2/1/2020 0:00	1658	14	Pacific	1494	1	440	THEFT PLAIN
11/8/2020 0:00	730	18	Southeast	1844	2	626	INTIMATE PAF
3/20/2020 0:00	1320	17	Devonshire	1752	2	624	BATTERY - SIM
8/17/2020 0:00	400	15	N Hollywo	1549	1	310	BURGLARY
1/19/2020 0:00	1850	14	Pacific	1435	1	310	BURGLARY
11/1/2020 0:00	1800	13	Newton	1313	1	440	THEFT PLAIN
1/20/2020 0:00	1540	14	Pacific	1494	1	440	THEFT PLAIN

Figure 1: CSV File with Date and Crime Code

Task 3 - Question 2 Analysis

How does crime vary within different areas across Los Angeles

Within the dataset, information about the date of the incident, area code, and the name of the area in which the area occurred are all present.

From the analysis, we will compare the total amount of crimes in different areas. We expect to find the relationship between certain areas and the average amount of crime over a period.

DATE OCC	TIME OCC	AREA	AREA NAME
11/7/2020 0:00	845	15	N Hollywood
10/18/2020 0:00	1845	15	N Hollywood
10/30/2020 0:00	1240	9	Van Nuys
12/24/2020 0:00	1310	7	Wilshire
9/29/2020 0:00	1830	14	Pacific
11/11/2020 0:00	1210	4	Hollenbeck
4/16/2020 0:00	1350	3	Southwest
7/7/2020 0:00	1400	11	Northeast
3/2/2020 0:00	1200	17	Devonshire
9/1/2020 0:00	900	21	Topanga
6/20/2020 0:00	1	6	Hollywood
11/17/2020 0:00	1320	20	Olympic
9/5/2020 0:00	1500	21	Topanga

Figure 2: CSV File with Date, Area Code, and Area Name

Task 3 - Question 3 Analysis

Crime vs Time of Day

As shown, data is provided for the exact time and date that the incident occurred. We can use this data to separate data into “night” or “day” categories and analyze the amount of data that occurs at each time or on specific days of the week.

From the analysis of this data, we will be able to analyze the relationship between what specific times of day have the most amount of crime, which will be useful in properly planning the amount of officers on duty. This question will allow us to look at more than just a specific time, and instead look at patterns or trends throughout the day to find peak hours.

DATE OCC	TIME OCC	AREA	AREA NAME	CrimeType	CrimeType Desc	VictAge	VictSex	VictDesc	WeaponDesc
11/7/2020 0:00	845	15	N Hollywood		354 THEFT OF IDE	31	M	H	
10/18/2020 0:00	1845	15	N Hollywood		230 ASSAULT WIT	32	M	H	KNIFE WITH BLA
10/30/2020 0:00	1240	9	Van Nuys		354 THEFT OF IDE	30	M	W	
12/24/2020 0:00	1310	7	Wilshire		331 THEFT FROM	47	F	A	
9/29/2020 0:00	1830	14	Pacific		420 THEFT FROM	63	M	H	
11/11/2020 0:00	1210	4	Hollenbeck		354 THEFT OF IDE	35	M	B	
4/16/2020 0:00	1350	3	Southwest		354 THEFT OF IDE	21	F	B	
7/7/2020 0:00	1400	11	Northeast		812 CRM AGNST C	14	F	H	UNKNOWN WEA
3/2/2020 0:00	1200	17	Devonshire		354 THEFT OF IDE	43	M	W	
9/1/2020 0:00	900	21	Topanga		354 THEFT OF IDE	57	M	W	
6/20/2020 0:00	1	6	Hollywood		812 CRM AGNST C	13	M	H	STRONG-ARM (H
11/17/2020 0:00	1320	20	Olympic		354 THEFT OF IDE	34	F	A	
9/5/2020 0:00	1500	21	Topanga		510 VEHICLE - ST	0			
7/2/2020 0:00	500	17	Devonshire		310 BURGLARY	0	X	X	
2/13/2020 0:00	2300	14	Pacific		330 BURGLARY FF	32	M	W	
2/1/2020 0:00	1658	14	Pacific		440 THEFT PLAIN	39	M	O	
11/8/2020 0:00	730	18	Southeast		626 INTIMATE PAF	26	F	B	STRONG-ARM (H
3/20/2020 0:00	1320	17	Devonshire		624 BATTERY - SIN	21	F	W	UNKNOWN WEA
8/17/2020 0:00	400	15	N Hollywood		310 BURGLARY	0	M	W	
1/19/2020 0:00	1850	14	Pacific		310 BURGLARY	37	M	A	UNKNOWN WEA
11/1/2020 0:00	1800	13	Newton		440 THEFT PLAIN	37	F	H	
1/20/2020 0:00	1540	14	Pacific		440 THEFT PLAIN	34	M	W	
1/29/2020 0:00	545	20	Olympic		440 THEFT PLAIN	0	X	X	
11/25/2020 0:00	1200	17	Devonshire		310 BURGLARY	69	F	W	
4/28/2020 0:00	1730	16	Foothill		230 ASSAULT WIT	14	M	H	AIR PISTOL/REV
2/11/2020 0:00	1600	20	Olympic		745 VANDALISM -	24	M	H	
5/13/2020 0:00	900	20	Olympic		520 VEHICLE - AT	36	M	H	
10/15/2020 0:00	305	19	Mission		740 VANDALISM -	35	F	H	

Figure 3: CSV File with Date, Time, and Type of Crime

Task 4 - Question 1, Metrics Tools and Plots

How has the total and frequency of different types of crimes changed over time from 2020 to 2024?

- Metrics:
 - Total crimes per year (2020–2023)
 - 2-Year Moving Average
 - Average Daily Crimes (mean)
 - Standard Deviation
 - T-Test comparing 2020 vs 2023
- Tools:
 - Pandas (data extraction / grouping)
 - Matplotlib (plotting)
 - Script.stats (stats tests)
- Plots:
 - Total Crimes per Year (line graph)
 - Moving Average (line plot)
 - Multi-line plot with Top 5 crimes

Task 4 - Question 2, Metrics Tools and Plots

How does the type of crime vary across different areas or neighborhoods in Los Angeles?

- Metrics:
 - Total crimes per area
 - Percentage distribution
- Tools:
 - Pandas (for value counts)
 - Matplotlib for graphs
- Plots:
 - Top 10 Areas of total crime in LA (bar chart)
 - Printed percent distribution calculations

Task 4 Question 3, Metrics Tools and Plots

How does the time of day influence the frequency and type of crimes committed?

- Metrics:
 - Number of crimes per hour
 - Peak crime hour
- Tools:
 - Pandas (time parsing & grouping)
 - Datetime functions
 - Matplotlib
- Plots:
 - Total crimes by hour (bar chart)

Dataset Errors

- Any event recorded without a time was defaulted to 12pm.
- The 2024 data was incomplete as the LAPD adopted a new records management system in March of 2024.

MONTH	
1	18363
2	16900
3	15795
4	12560
5	9161
6	7876
7	7967
8	8114
9	8104
10	7605
11	6977
12	4605

Figure 4: Monthly Reports Throughout 2024

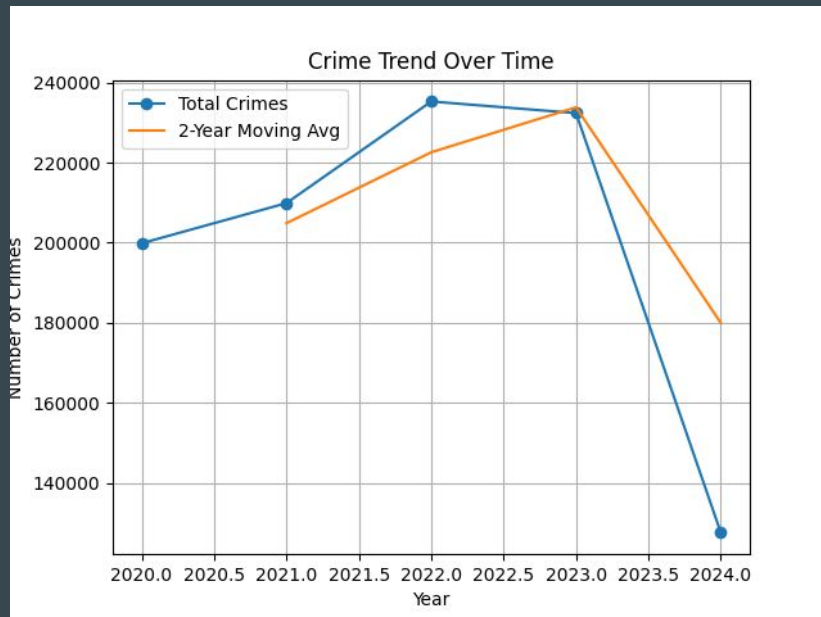


Figure 5: Unfiltered Total Crime Data

Task 5 Q1 - Output Results

Crimes per Year

Year	Count
2020	199,847
2021	209,876
2022	235,259
2023	232,345

Yearly Growth Rate (%)

Year	Growth %
2021	5.018%
2022	12.094%
2023	-1.239%

Daily Statistics

Metric	2020	2023
Avg Daily Crimes	546.03	636.56
Std Deviation	69.05	72.93

Top Crime Types

Rank	Crime Type
1	Stolen Vehicle
2	Battery - Assault
3	Identity Theft
4	Burglary from Vehicle
5	Burglary

Table 1: Total and Types of Crime Analysis

Task 4 Q1 - Resulting Graphs

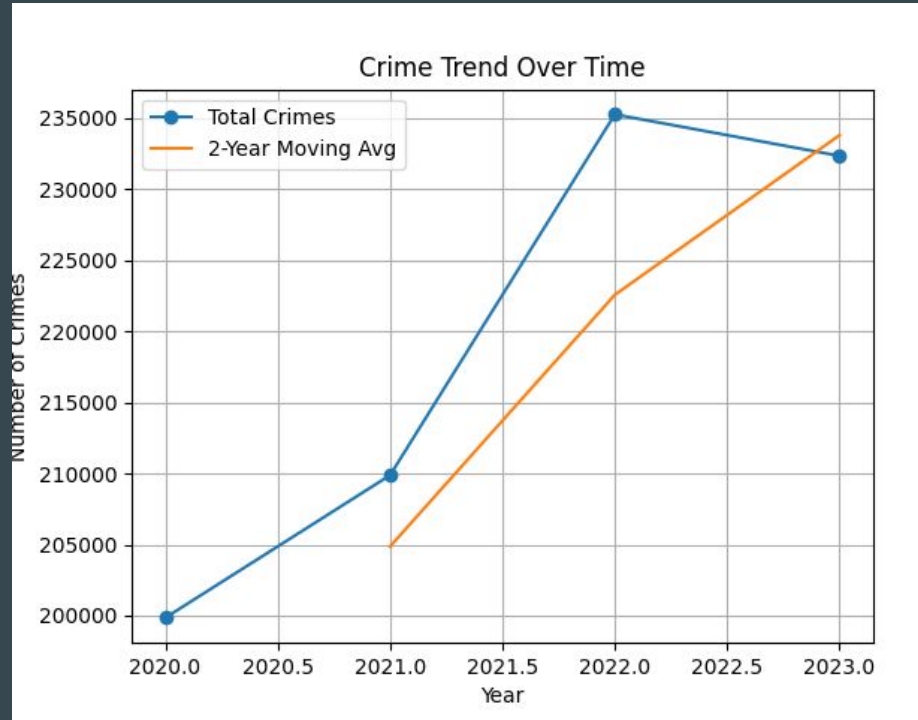


Figure 6: Total Crime per year

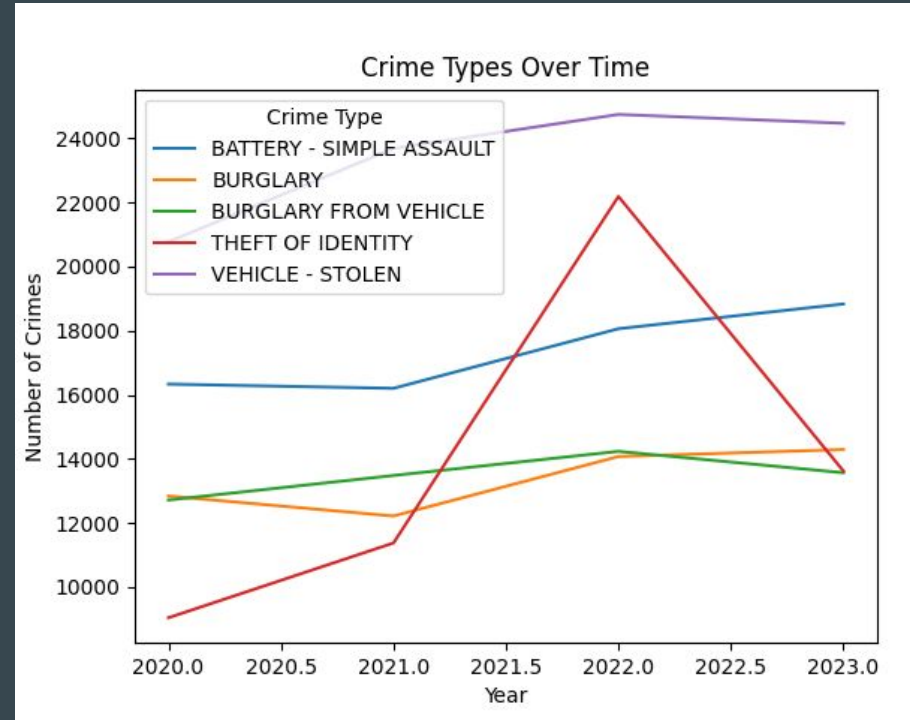


Figure 7: Top 5 Crimes over Time

Task 5 Q2 - Output Results

Division	Crimes
Central	59,456
77th Street	54,981
Pacific	51,351
Southwest	49,127
Hollywood	46,277
Southeast	44,325
Olympic	44,300

Division	Crimes
N Hollywood	43,858
Newton	43,787
Wilshire	41,933
Rampart	41,223
West LA	40,313
Northeast	37,703
Van Nuys	37,122

Division	Crimes
West Valley	36,864
Harbor	36,085
Devonshire	35,864
Topanga	35,826
Mission	34,982
Hollenbeck	32,768
Foothill	29,182

Table 2: Total Crimes in each area of Los Angeles

Task 5 Q2 - Plots

These results line up with the expected results as Central LA and 77th street are known for their high gang prevalence.

[Article about 77th Street Violence](#)

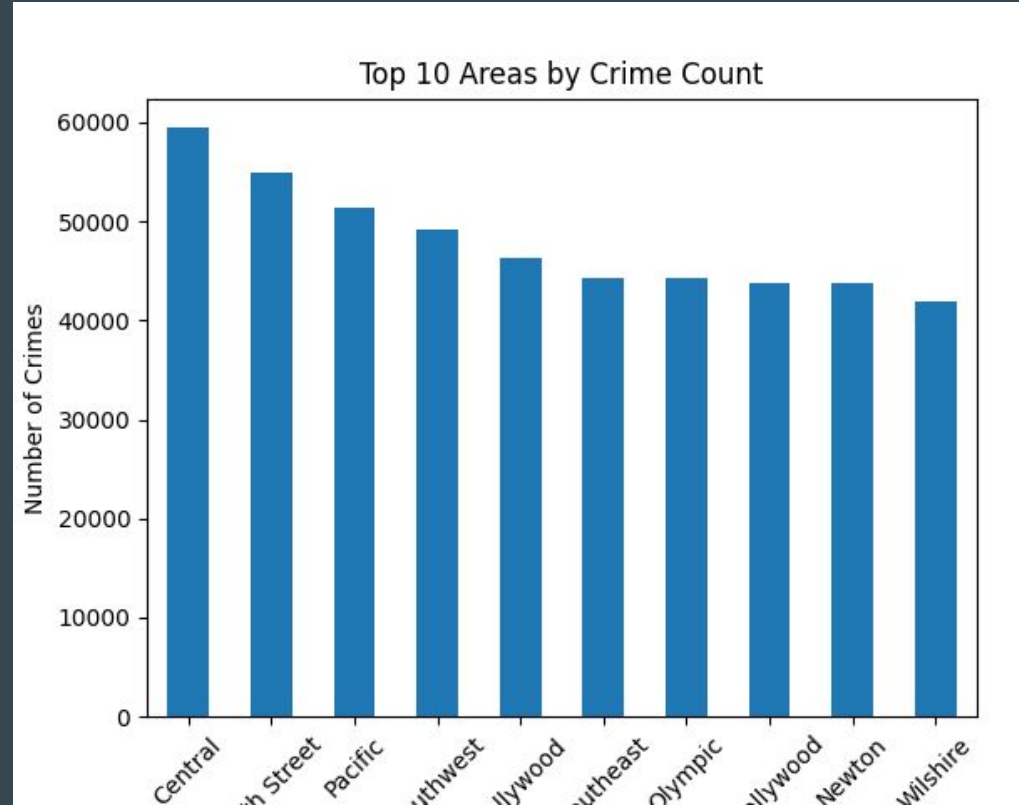


Figure 8: Top 10 Areas of Crime in LA

Task 5 Q3 - Results

Crimes per Hour

Hour	Crimes
0	36,280
1	26,390
2	22,350
3	19,531
4	16,501
5	15,257
6	19,997
7	22,618

Hour	Crimes
8	32,483
9	31,848
10	37,707
11	38,173
12	28,324
13	39,708
14	42,836
15	45,760

Hour	Crimes
16	46,064
17	50,975
18	52,061
19	48,030
20	48,772
21	44,386
22	42,754
23	36,864

Peak Hour: 18 (52,061 crimes)

Table 3: Total Crimes at Each Hour of the Day

Task 5 Q3 - Graphs

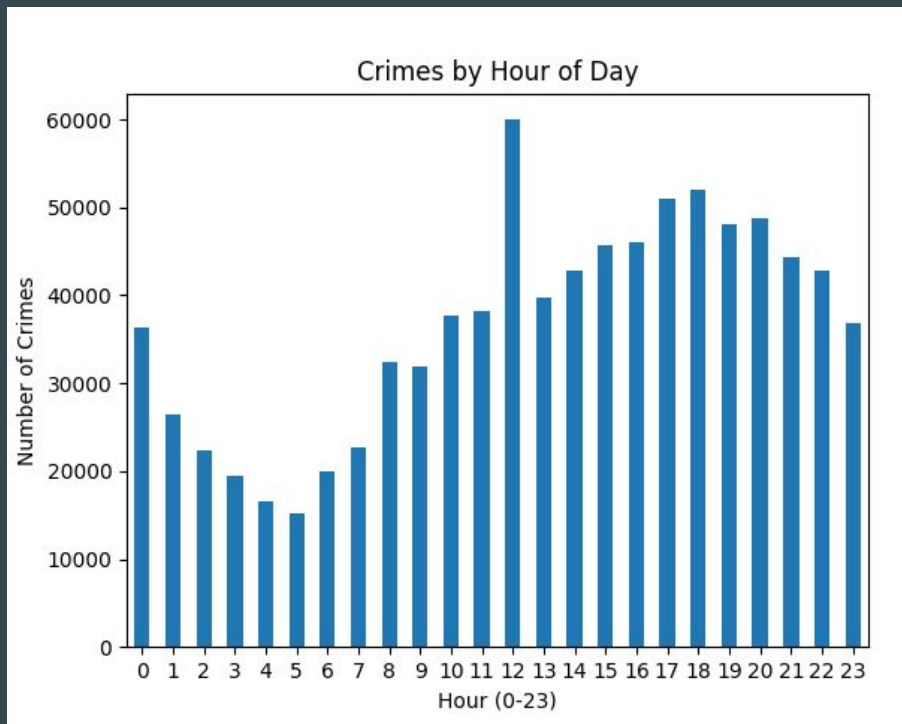


Figure 9: Unfiltered Total Crime per Hour

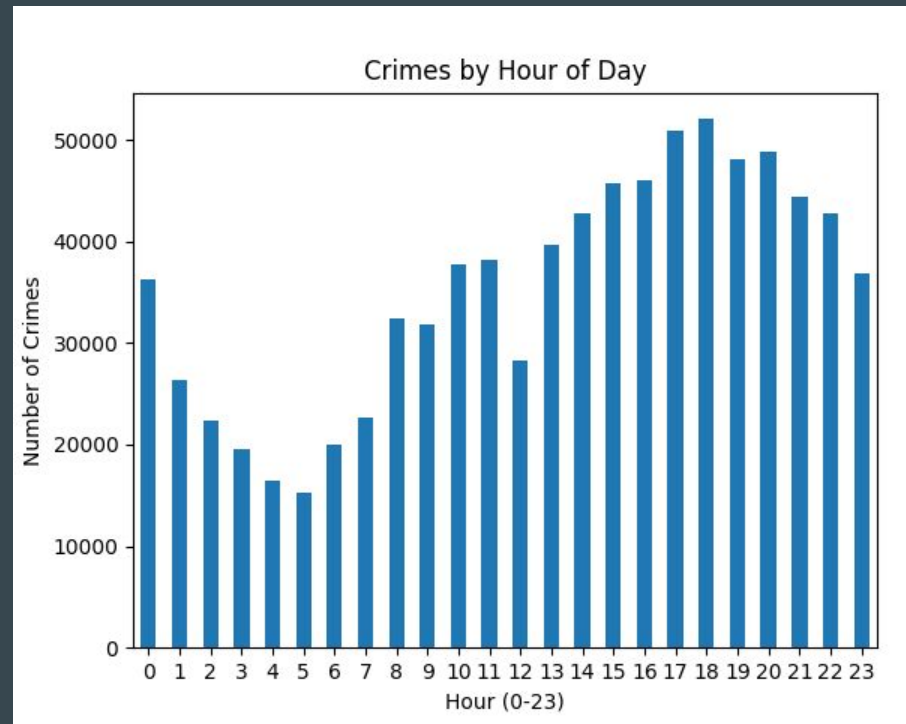


Figure 10: Filtered Total Crime per Hour

Task 6 – Question 1 (Crime Over Time)

- Conclusion:
 - There was a drop in crime during the COVID-19 pandemic, which appears as an increase in crime in the following years
- Did results match expectations?
 - For total crime, Yes. We expected the total amount of crime to be down during those years as the pandemic kept lots of people isolated and showed a clear trend in our findings.
 - As for specific types of crimes, there was no clear trend as each crime followed the general overall increase. To better answer the question, analyzing the total percentage of each type of crime over time would demonstrate a better indication of trends.
- Limitations:
 - Does not account for population changes or reporting differences.
 - The dataset was smaller than expected, as the 2024 results were unfinished

Task 6 – Question 2 (Crime by Area)

- Conclusion:
 - The crime distribution in LA is similar to other cities where certain areas are more dangerous than others. Crime is spread out through LA, but certain areas have much higher levels of crime such as in Central LA and 77th Street.
- Did results match expectations?
 - Yes, the results match expectations and demonstrated a clear trend as areas known for many surrounding gangs, such as 77th street, had very high levels of crime, whereas neighborhoods known for being quiet and tree-lined, such as the Foothill area, have much lower rates of crimes.
- Limitations:
 - The dataset does not consider population density or socioeconomic factors.

Task 6 – Question 3 (Time of Day)

- Conclusion:
 - Crime frequency follows an oscillating pattern with crime being highest around 6pm, and being lowest at 5am.
- Did results match expectations?
 - Yes, this pattern makes sense as crime tends to increase during evening hours, and would be lowest during sleeping hours. However, we expected this data to be shifted slightly right with crime peaking closer to 9pm.
- Justification:
 - The bar chart demonstrates a clear trend throughout the day, and aligns with the expected hours of higher or lower crime.
- Limitations:
 - Does not account for day of week or special holidays

Task 7 - Reflection

- What we learned:
 - Additional research may be important as it provides additional insight to the reason behind certain trends
 - How to and the importance of cleaning large datasets
 - How visualization provides an additional level of insight to data interpretation
- What surprised us:
 - How much preprocessing data required before analysis
 - Some trends were not as obvious as expected
 - Small decisions (grouping, filtering) impacted results significantly
- Thoughts on the process:
 - Helped improve our problem-solving and analytical thinking
 - Some unfiltered datasets or alternative questions may be require much more intensive filtering and